

Group Assessment Coversheet

To be attached to the front of the assessment.

Campus: **Tyger Valley**

Faculty: **Information Technology**

Module Code: **ITNSA2-11**

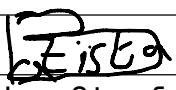
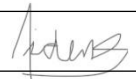
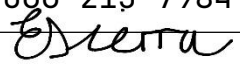
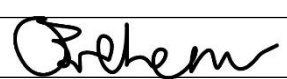
Group: **7**

Lecturer's Name: **Amakan Agoni**

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Table of Contents

Section A

Question 1.....Page 1-7

Question 2..... Page 8-10

Question 3.....Page 11-12

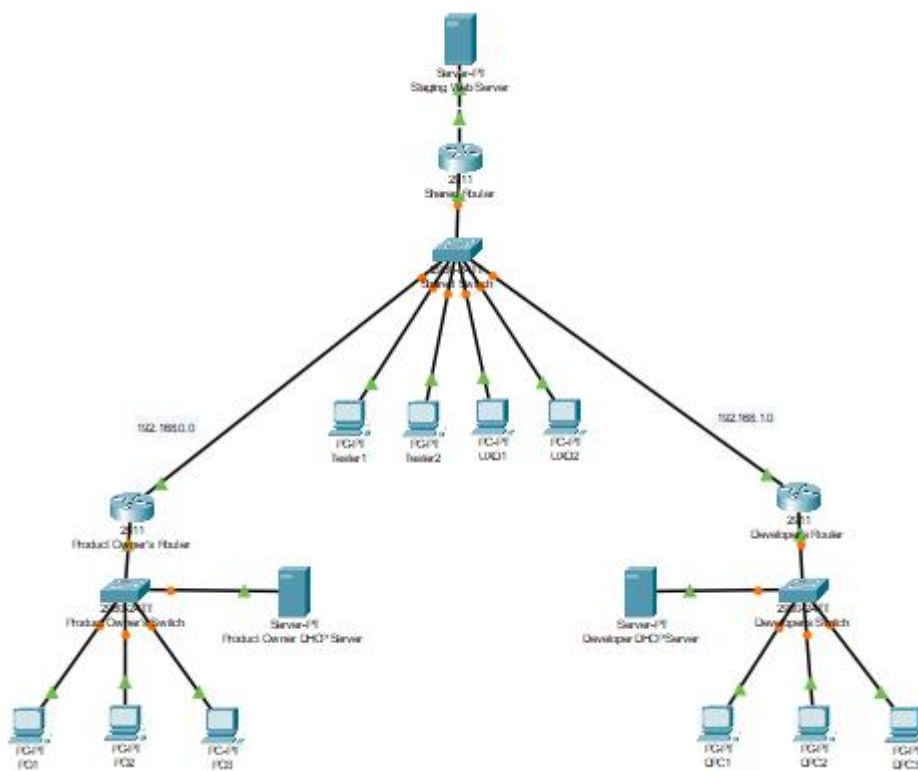
Question 4..... Page 13-15

Bibliography..... Page 16

Section A

Question 1

1.1) We designed a network layout in Cisco Packet Tracer that reflects the three floors of the company's infrastructure through star topologies connected together by a central switch. On floor one, there are 3 devices for the development team, 3 devices for the product owners, and their routers and switches. On floor two, 2 devices are placed for the testers and 3 devices for the UX developers. Finally, on floor 3, a staging web server is placed. A shared router is also included to handle the network's traffic for both floor 2 and 3. Switches were added to connect the devices on each floor. The topology provides a structured VLAN-based network, which enables BusyBugs to easily scale the network up or down as needed while maintaining efficient traffic flow.



1.2) When setting up DHCP for devices in the Development and Product Owner networks, the routers for each router must be enabled and assigned an IP address that allows for the DHCP to communicate and assign IP addresses with the other devices in the networks.

Development router:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
```

Product Owner router:

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.0.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
```

The IP address, subnet mask and default gateway (corresponding router's IP address) for each DHCP server must be configured to allow for it to communicate through the router and assign IP addresses.

Development DHCP server:

IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	0.0.0.0

Product Owner DHCP server:

IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.0.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.0.1
DNS Server	0.0.0.0

DHCP must then be enabled and configured on each DHCP server that allows it to assign IP addresses while meeting requirements and limitations that are specified. These requirements and limitations includes specifying the default gateway, subnet mask, starting IP address and max number of users.

Development DHCP server:

DHCP

Interface FastEthernet0 Service ☒ On ☐ Off

Pool Name serverPool

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

Start IP Address : 192 168 1 0

Subnet Mask: 255 255 255 0

Maximum Number of Users : 256

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168....	0.0.0.0	192.168....	255.255....	256	0.0.0.0	0.0.0.0

Product Owner DHCP server:

DHCP

Interface FastEthernet0 Service ☒ On ☐ Off

Pool Name serverPool

Default Gateway 192.168.0.1

DNS Server 0.0.0.0

Start IP Address : 192 168 0 0

Subnet Mask: 255 255 255 0

Maximum Number of Users : 256

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168....	0.0.0.0	192.168....	255.255....	256	0.0.0.0	0.0.0.0

Each PC can then be configured to have a DHCP IP address instead of a static IP address.

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

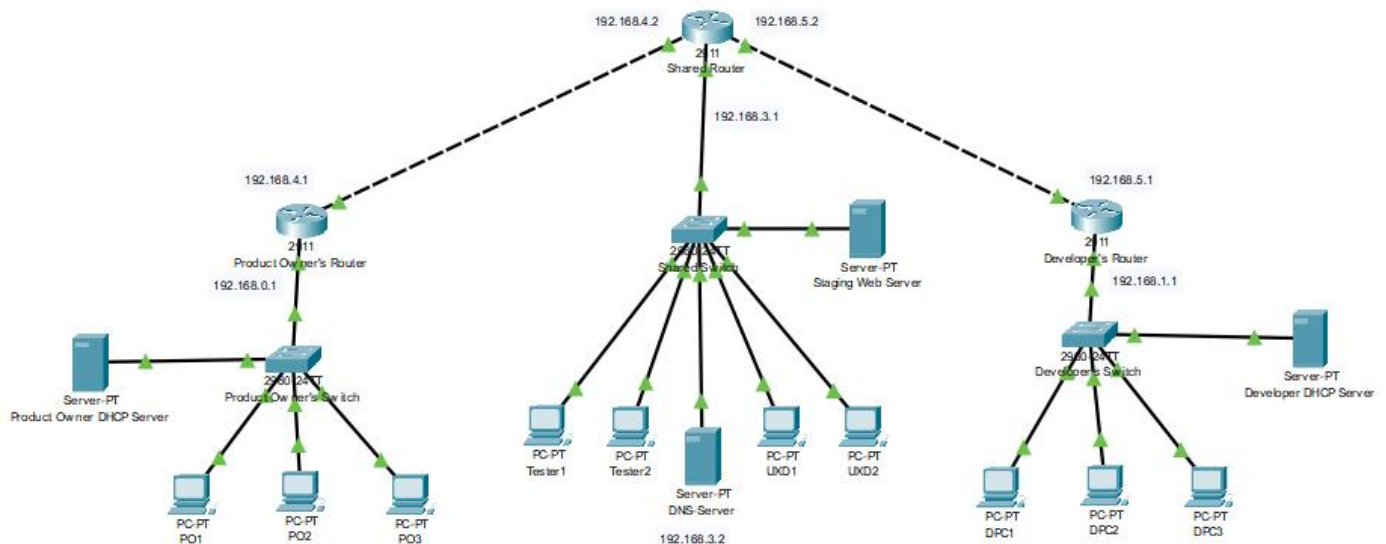
IPv4 Address: 192.168.0.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.0.1

DNS Server: 0.0.0.0

1.3) An issue occurred when implementing the DNS server as the devices in the Development and Product Owner networks were not able to reach the server. The layout of the whole network was altered to enable these devices to reach the DNS server. The network was changed by connecting all the routers to each other instead of utilising a switch to connect them to each other.



Static routing is enabled on each router to temporarily provide routing between each sub network and allow devices to communicate over the whole network.

Development router:

Static Routes

Network	
Mask	
Next Hop	
Add	

Network Address

192.168.4.0/24 via 192.168.5.2

192.168.3.0/24 via 192.168.5.2

Shared router:

Static Routes

Network	
Mask	
Next Hop	
Add	

Network Address

192.168.0.0/24 via 192.168.4.1

192.168.1.0/24 via 192.168.5.1

Product Owner router:

Static Routes

Network	
Mask	
Next Hop	
Add	

Network Address

192.168.3.0/24 via 192.168.4.2

192.168.5.0/24 via 192.168.4.2

A DNS server is added to the to the floor two network that enables all devices within each aspect of the whole BusyBugs network. After the DNS server is added, it must be configured to have an IP address, subnet mask and default gateway in order to communicate in the network.

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.3.2

Subnet Mask

255.255.255.0

Default Gateway

192.168.3.1

DNS Server

0.0.0.0

The DNS service must then be enabled and configured that provides names to each tester and UX developers devices. It's configured by specifying it with a name and a corresponding IP address.

DNS

DNS Service

☒ On

☐ Off

Resource Records

Name

tester1.com

Type

A Record

Address

192.168.10.2

Add

Save

Remove

No.	Name	Type	Detail
0	tester1.com	A Record	192.168.10.2
1	tester2.com	A Record	192.168.10.3
2	uxdeveloper1.com	A Record	192.168.20.2
3	uxdeveloper2.com	A Record	192.168.20.3

The DHCP servers must then be reconfigured to provide the IP address of the DNS server to each device in the network.

Development DHCP server:

DHCP

Interface

FastEthernet0

Service

☒ On

☐ Off

Pool Name

serverPool

Default Gateway

192.168.1.1

DNS Server

192.168.3.2

Start IP Address :

192

168

1

0

Subnet Mask:

255

255

255

0

Maximum Number of Users :

256

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168....	192.168....	192.168....	255.255....	256	0.0.0.0	0.0.0.0

Product Owner DHCP server:

DHCP

Interface FastEthernet0 Service ☒ On ☐ Off

Pool Name serverPool

Default Gateway 192.168.0.1

DNS Server 192.168.3.2

Start IP Address : 192 168 0 0

Subnet Mask: 255 255 255 0

Maximum Number of Users : 256

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168....	192.168....	192.168....	255.255....	256	0.0.0.0	0.0.0.0

To ensure that the DNS server is now reachable by the devices in the network, we utilised the ping feature to determine the connectivity between the device and the DNS server.

PC from the Product Owner sub network:

```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Question 2

2.1) Unique VLANs should be created in the switch for the testers and UX developers. VLAN 10 was assigned for the testers and VLAN 20 for the UX developers.

```
Switch(config)#vlan 10
Switch(config-vlan)#name Tester
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name UX
Switch(config-vlan)#exit
```

Each port of the switch that has an end user device connected to it must be configured in access mode and be assigned with the appropriate VLAN that the device belongs to. The port of the switch that the router is connected to is configured differently to the end user devices where the port is configured into trunk mode.

```
Switch(config)#int fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#int fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit

Switch(config)#int gig0/1
Switch(config-if)#switchport mode trunk
```

2.2) The router must then be configured where each VLAN is identified by the router and each VLAN will be assigned a default gateway that enables communication between the VLANs.

```
Router(config)#int gig0/0.10
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit

Router(config)#int gig0/0.20
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
```

We then ping the devices from each VLAN to test if it's reachable.

A PC from VLAN 10 pings a PC from VLAN 20:

```
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The static routing between the router was then modified to allow devices in the other sub networks to communicate in the VLAN.

Development router:

Static Routes

Network	
Mask	
Next Hop	
Add	

Network Address
192.168.4.0/24 via 192.168.5.2
192.168.3.0/24 via 192.168.5.2
192.168.10.0/24 via 192.168.4.2
192.168.20.0/24 via 192.168.4.2
Remove

Product Owner router:

Static Routes

Network	
Mask	
Next Hop	
Add	

Network Address
192.168.3.0/24 via 192.168.4.2
192.168.5.0/24 via 192.168.4.2
192.168.10.0/24 via 192.168.4.2
192.168.20.0/24 via 192.168.4.2
Remove

2.3) An ACL is created in the shared router to deny traffic between the VLANs. Traffic from VLAN 10 (Testers) to VLAN 10 (UX Developers) is denied as well as all traffic from VLAN 10 (UX Developers) to VLAN 10 (Testers). All other traffic is allowed as needed.

```
Router(config)#access-list 102 deny ip host 192.168.10.1 192.168.20.0 0.0.0.255
Router(config)#access-list 102 deny ip host 192.168.20.1 192.168.10.0 0.0.0.255
```

Access-group 102 in was applied on GigabitEthernet0/0.10 and GigabitEthernet0/0.20 to enforce restrictions.

```
Router(config)#int gig0/0.10
Router(config-subif)#ip access-group 102 in
Router(config-subif)#exit
Router(config)#int gig0/0.20
Router(config-subif)#ip access-group 102 in
Router(config-subif)#exit
```

Ping is used to test that the ACL works properly.

PC from VLAN 10 pinging a PC from VLAN 20.

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```


Question 3

3.1) In order to allow external partners to be able to reach the staging web server on floor 3 for demos while still restricting unauthorized access to the rest of the network, a NAT must be configured. This was done by assigning a public IP addresses to the staging web server to allow for external access, a static NAT is used to map an external public IP address to protect the staging web's server's private IP.

```
Router(config)#interface gigabitethernet0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#ip nat pool shared_pool 192.168.100.1 192.168.100.101 netmask
255.255.255.0
Router(config)#ip nat inside source list 1 pool shared_pool
```

(ManageEngine Network Configuration Manager, 2024)

Next, an ACL is set up to restrict unwanted access. This was done by creating an ACL to only allow approved external partner IPs to access the staging server. Other connections are blocked while internal development teams are still permitted.

```
Router(config)#access-list 101 deny ip host 192.168.3.3 any
Router(config)#access-list 101 permit tcp host 192.168.3.3 any eq 80
Router(config)#access-list 101 permit tcp host 192.168.3.3 any eq 443
```

3.2)

1. Network Segmentation

1. Dedicated VLAN for Staging Environment

- Create a separate VLAN for the staging web server and related services.
- Assign it a subnet.

2. Firewall Segmentation

- Place the staging environment behind a firewall or a router ACL to restrict unnecessary access.
- Use Access Control Lists (ACLs) to allow only HTTP/HTTPS traffic from external users.

3. Separate Subnets for Internal Users

- Developers: 192.168.2.0/24 (VLAN 20)
- Testers: 192.168.10.0/24 (VLAN 10)
- Other internal users should be isolated from the staging VLAN unless required.

2. External Access & Security Controls

1. Public IP Mapping (NAT/PAT)

- Assign a public IP that maps to the staging server's private IP via Port Address Translation (PAT).

- Forward only port 80 (HTTP) and 443 (HTTPS).

2. Firewall & ACL Rules

- Allow external users to access the staging environment via HTTP/HTTPS.
- Block all other inbound access except for specific maintenance tasks.
- Restrict SSH/management access to only the internal network.

3. Load Balancing & Redundancy

- If needed, implement a reverse proxy or load balancer to distribute traffic.

3. Internal Access & DevOps Integration

1. Developer & QA Access

- Developers and Testers should access the staging environment only from the internal network.
- Use SSH, RDP, or VPN for secure remote management.

2. Logging & Monitoring

- Enable Syslog logging, Intrusion Detection (IDS), and real-time monitoring for external access attempts.

3. Automated Deployments

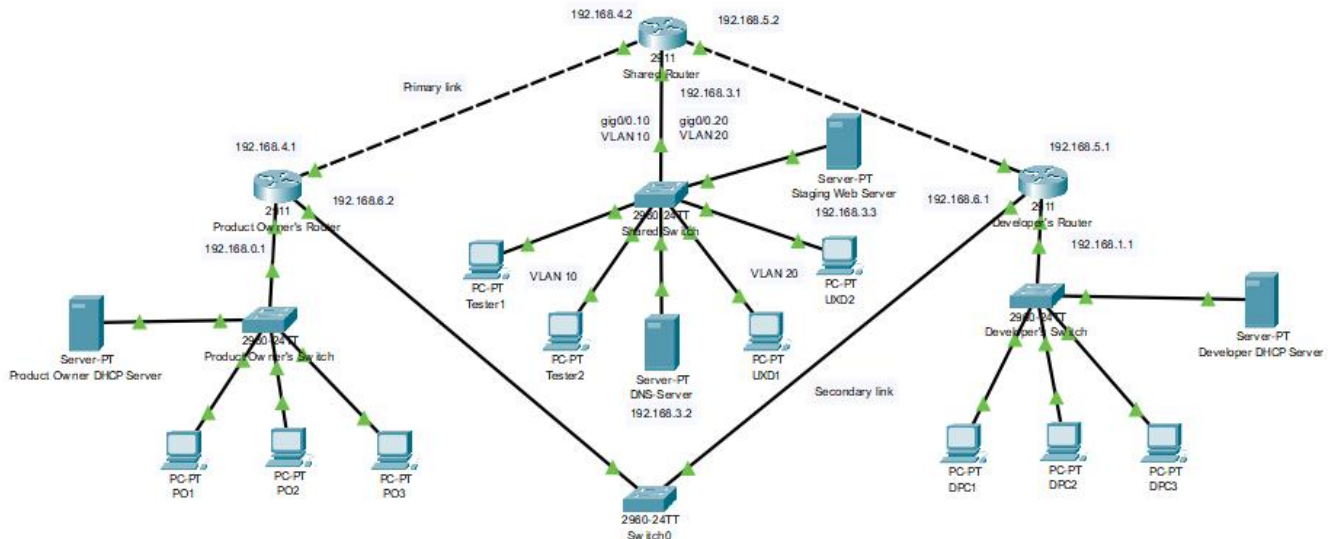
- Implement CI/CD pipelines to deploy updates from development to staging before pushing to production.

4. Summary of Theoretical Design

<i>Component</i>	<i>Implementation</i>
<i>VLANs</i>	Separate VLAN for staging (VLAN 20)
<i>Firewall and ACLs</i>	Restrict inbound traffic to HTTP/HTTPS only
<i>NAT/PAT</i>	Map public IP to internal staging server
<i>Internal Access</i>	Developers/Testers access only via internal network
<i>Monitoring</i>	Enable logging, IDS, and real-time alerts

Question 4

4.1) To enable a failover feature for the network, a switch that connects between the development and product owner routers is implemented due to the “sla” command being not available for the shared router. This switch enables the shared router to take an additional route if another route to one of the routers failed.



RIP is configured on the development router that enables routing between the networks 192.168.1.0, 192.168.5.0 and the new network between the development and product owner routers 192.168.6.0. RIP is enabled specifically on this router as the developers can make use of this legacy feature in the future.

```
Router(config)#route rip
Router(config-router)#version 2
Router(config-router)#network 192.168.1.0
Router(config-router)#network 192.168.5.0
Router(config-router)#network 192.168.6.0
Router(config-router)#no auto-summary
```

((Sal), 2019)

OSPF should be configured on the shared router to improve the efficiency of routing to the staging environment.

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.3.1 0.0.0.255 area 0
Router(config-router)#network 192.168.4.2 0.0.0.255 area 0
Router(config-router)#network 192.168.5.2 0.0.0.255 area 0
```

(Academy, 2022)

A failover is then introduced for the development and product owner routers by adding an additional route where the shared router can connect to one of the routers with the other one.

```
Router(config)#ip route 192.168.0.0 255.255.255.0 192.168.4.1
Router(config)#ip route 192.168.0.0 255.255.255.0 192.168.6.2 10
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.5.1
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.6.1 10
```

(Maupin, 2013)

4.2) Running both RIP (Routing Information Protocol) and OSPF (Open Shortest Path First) in the same network introduces several challenges, especially when implementing failover or backup links for resilience.

1. Routing Protocol Differences

- RIP is a protocol that uses hop count for determining the best routing path.
- OSPF is a protocol that determines the best routing path through aspects like the delay when routing.
- OSPF converges faster and supports larger networks, while RIP is slower and more prone to loops.

RIP's simplicity can cause suboptimal routing decisions when interacting with OSPF that can lead to inefficient path selection.

2. Redistribution Complexity

- When running both protocols, route redistribution is required to exchange routes between RIP and OSPF domains. Redistribution can lead to route loops or suboptimal routing paths.

Properly setting metrics and administrative distance is crucial to avoid loops and ensure the primary link is preferred while the backup link remains idle until a failure occurs.

3. Administrative Distance Conflict

- By default, OSPF has a lower administrative distance (110) than RIP (120), meaning OSPF routes are always preferred. This can lead to situations where RIP routes are ignored, even when they might be more appropriate for backup paths.

Adjusting administrative distance manually can help, but it adds complexity and requires careful tuning.

4. Convergence Time Differences

- RIP's slow convergence time (up to 180 seconds) can delay failover to backup links.
- OSPF converges much faster due to triggered updates and topology awareness.

During a failover scenario, the network may experience routing black holes or loops due to RIP's delay.

5. Inconsistent Network Topology Views

- OSPF builds a complete link-state database, while RIP only knows about directly connected routes and next hops. This mismatch can lead to routing inconsistencies when redistributing routes between protocols.

Carefully filtering routes during redistribution is necessary to prevent suboptimal routes from propagating.

6. Scalability Concerns

- OSPF is designed for large, hierarchical networks, while RIP is limited to smaller environments. Mixing them can create scalability issues, especially if the network grows.

Recommended Solutions

1. Use route filtering and tagging to prevent routing loops.
2. Set administrative distance carefully to prioritize OSPF routes while allowing RIP as a backup.
3. Implement route summarization to reduce the number of routes being redistributed.
4. Monitor and test failover scenarios to ensure smooth transitions between primary and backup links.

(Paluch, 2011)

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